

Measurement Strategies in Quantum Computing

Why Measuring All Qubits is Optimal in Modern Algorithms

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Central question

Why is it preferable to measure all qubits rather than mapping observables onto a single qubit?

- Traditional textbooks emphasize single-qubit readout.
- Modern quantum algorithms (VQE, QML) rely on full-register measurement.

Quantum measurement structure

An n -qubit state is

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle.$$

Measurement in the computational basis yields:

$$p(x) = |c_x|^2.$$

Key observation

The outcome is a **joint probability distribution** over all qubits.

Theorem: Information loss in partial measurement

Theorem

Let ρ be an n -qubit state. Measuring only a subset of qubits yields marginal distributions which do not uniquely determine ρ or its multi-qubit correlations.

Proof.

Let ρ be entangled:

$$\rho \neq \rho_1 \otimes \rho_2.$$

Then in general:

$$\langle Z_i Z_j \rangle \neq \langle Z_i \rangle \langle Z_j \rangle.$$

Measuring only qubit i gives access to $\langle Z_i \rangle$, but not to joint correlators.

Hence the reduced statistics are insufficient to reconstruct expectation values of multi-qubit observables. □

Corollary: Necessity of full measurement

Corollary

To estimate expectation values of general observables,

$$H = \sum_k c_k P_k,$$

with P_k multi-qubit operators, one must measure all qubits involved.

$$\langle P_k \rangle = \langle Z_{i_1} Z_{i_2} \cdots X_{j_1} \rangle$$

- Requires joint measurement outcomes
- Cannot be reconstructed from single-qubit marginals

Theorem: Circuit-depth optimality

Theorem

Let O be a multi-qubit observable. Any protocol that maps O onto a single qubit prior to measurement requires a unitary transformation whose circuit depth is at least proportional to the support size of O .

Sketch.

Mapping O to a single qubit requires a unitary U such that

$$U^\dagger Z_1 U = O.$$

If O acts on k qubits, then U must entangle these qubits with qubit 1.

Thus:

- Requires at least $\mathcal{O}(k)$ entangling gates
- Increases circuit depth and noise

Direct measurement avoids this overhead.



Asymmetry in cost

- Quantum gates: noisy and expensive
- Measurements: comparatively cheap and parallelizable

Thus:

Adding gates (basis rotations, mappings) $\gg$ measuring more qubits

Deferred measurement principle

Principle

Measurements can be deferred until the end of a quantum circuit without changing outcome probabilities.

⇒ One can measure all qubits at the end

This supports the modern paradigm of full-register measurement.

Modern paradigm (NISQ era)

Variational algorithms

In hybrid algorithms:

prepare state $\rightarrow$ measure all qubits $\rightarrow$ classical post-processing

- Introduced with VQE (2014)
- Core workflow: sampling full bitstrings

Key question

Problem

Can we reconstruct expectation values of a Hamiltonian using only single-qubit measurements?

We show:

- This fails in general
- It fails even for simple two-qubit systems

Theorem: Failure of marginal reconstruction

Theorem

There exist quantum states ρ and observables H such that the expectation value $\langle H \rangle$ cannot be reconstructed from all single-qubit measurement statistics.

Proof.

Consider a two-qubit system and the observable

$$H = Z_1 Z_2.$$

Take two states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad \rho_{\text{mix}} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|).$$

Both have identical single-qubit marginals:

$$\langle Z_1 \rangle = \langle Z_2 \rangle = 0.$$

Explicit VQE counterexample

Consider Hamiltonian:

$$H = Z_1 Z_2.$$

Ansatz:

$$|\psi(\theta)\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle.$$

Exact energy:

$$E(\theta) = \langle Z_1 Z_2 \rangle = 1.$$

Single-qubit estimate:

$$\langle Z_1 \rangle = \cos^2 \theta - \sin^2 \theta, \quad \langle Z_2 \rangle = \cos^2 \theta - \sin^2 \theta.$$

Naive reconstruction:

$$\langle Z_1 Z_2 \rangle \approx \langle Z_1 \rangle \langle Z_2 \rangle = (\cos^2 \theta - \sin^2 \theta)^2.$$

Failure

This is **incorrect** except at special points.

Error in VQE energy landscape

True energy:

$$E_{\text{true}}(\theta) = 1.$$

Naive single-qubit estimate:

$$E_{\text{naive}}(\theta) = (\cos 2\theta)^2.$$

- True landscape: flat
- Naive landscape: oscillatory

Consequence

Optimization will converge to **wrong minima**.

Theorem: VQE failure under marginal approximation

Theorem

Replacing multi-qubit correlators with products of single-qubit expectations leads to incorrect energy landscapes and can produce false minima in VQE.

Proof.

Let $H = Z_1 Z_2$. True:

$$E(\theta) = \langle Z_1 Z_2 \rangle.$$

Approximation:

$$E'(\theta) = \langle Z_1 \rangle \langle Z_2 \rangle.$$

Since correlations are neglected:

$$E'(\theta) \neq E(\theta),$$

and the optimization problem changes. Thus the minimizer of E' is not the minimizer of E . □

Why textbooks differ

- Early algorithms (Deutsch–Jozsa, phase estimation)
 - Encode answer in a single qubit/register
- Analytical simplifications:
 - Easier to track one observable
- Phase kickback techniques:
 - Transfer information to one qubit

Conclusion

These are **pedagogical or theoretical constructions**, not optimal experimental strategies.

Final theorem: Optimal measurement strategy

Theorem

For variational and sampling-based quantum algorithms, measuring all qubits in the computational basis minimizes circuit depth while preserving full information about the quantum state.

Argument.

- Full measurement yields complete probability distribution $p(x)$
- Multi-qubit observables are computed from bitstrings
- Avoids additional unitary transformations
- Reduces noise accumulation

Thus, full-register measurement is optimal under realistic hardware constraints. □

Final conclusion

- Single-qubit measurements give only marginals
- Correlations are essential for quantum advantage
- VQE fails if correlations are ignored
- Full-register measurement is fundamentally necessary

Key insight

Quantum algorithms operate on **correlations**, not individual qubits.

Summary

- Quantum information is inherently **non-local**
- Single-qubit measurement loses correlations
- Mapping observables increases circuit depth
- Modern hardware supports parallel measurement
- NISQ algorithms rely on full bitstring sampling

Key insight

Quantum computing is fundamentally about sampling high-dimensional distributions, not extracting single bits.